

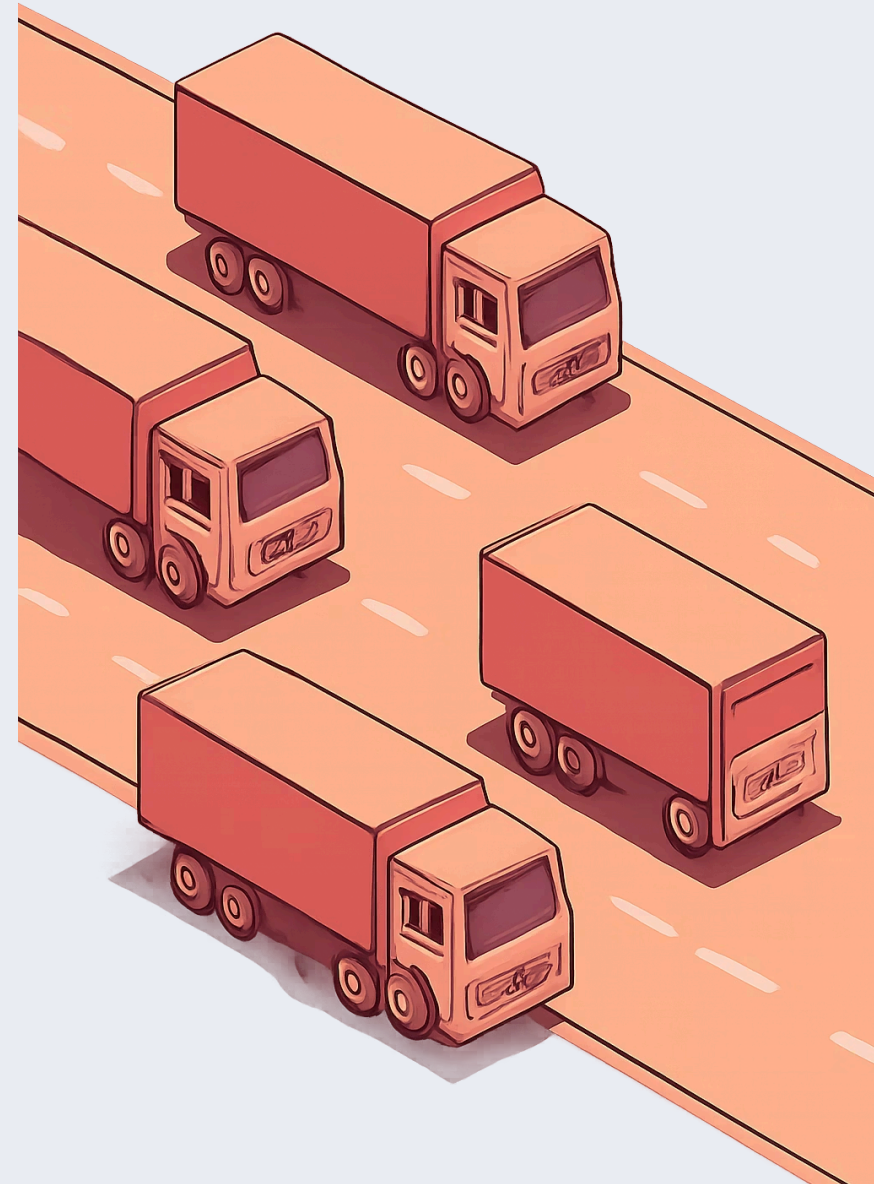
Fleet Management Dashboard

Transportation Domain

Name: G. KANIMOZHI

Department: BCA Sec "A"

Roll No: E25CA147



Problem & Solution Overview

Challenges in Fleet Management

Difficulty monitoring multiple vehicles simultaneously

Manual maintenance tracking and record-keeping

Fuel wastage and lack of consumption insights

Poor visibility into vehicle utilization

Lack of predictive maintenance capabilities

Our Solution

Centralized dashboard for real-time vehicle monitoring

Automated fuel and maintenance tracking

Live GPS tracking with interactive maps

ML-powered predictive maintenance

Comprehensive fleet analytics and reporting

Key Features & Dashboard



Real-Time Monitoring

Vehicle status, location, and performance metrics at a glance



Fuel Management

Consumption tracking, efficiency analysis, and cost optimization



Maintenance Alerts

Service reminders, overdue notifications, and priority levels



Trip Management

Route tracking, distance analytics, and complete trip history



Live Map Tracking

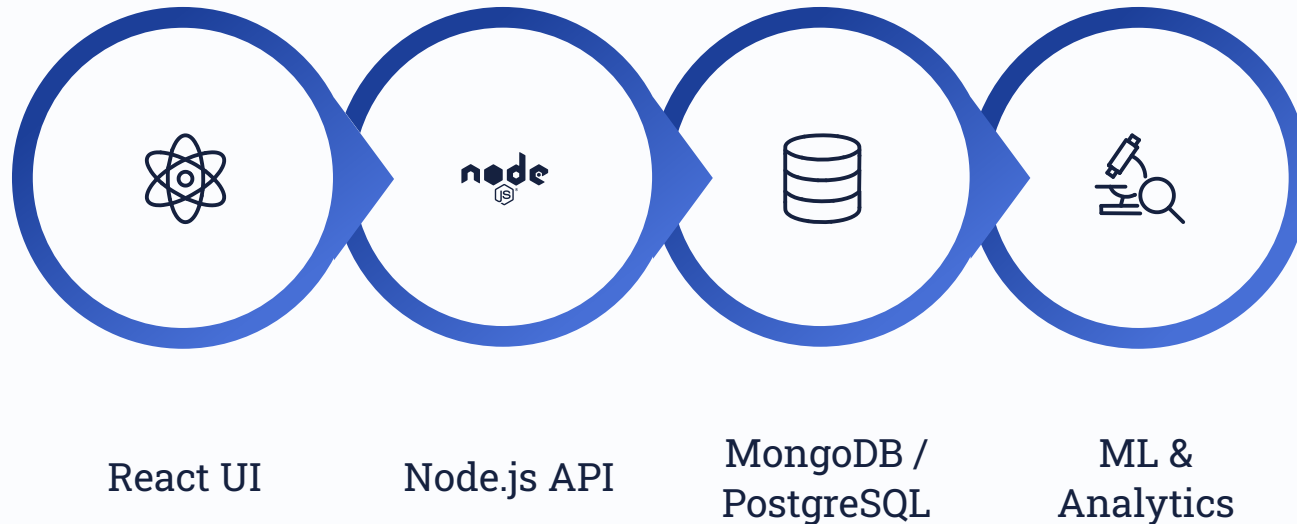
Interactive Leaflet map with vehicle status indicators



Fleet Analytics

Performance charts, utilization rates, and cost analysis

Technology Stack & Architecture



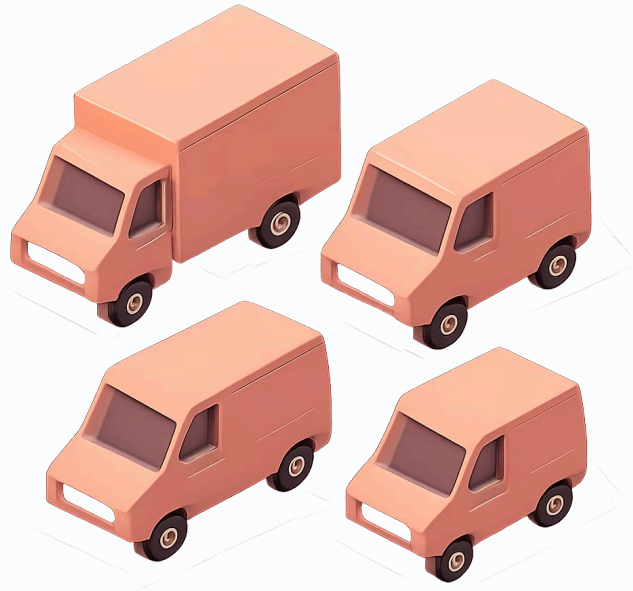
Frontend & Backend

- **Frontend:** React.js, Vite, CSS — Modern responsive UI
- **Backend:** Node.js, Express.js — Scalable API
- **Database:** MongoDB / PostgreSQL — Reliable data storage

Data Visualization & Tracking

- **Charts:** Recharts / Charts.js for interactive dashboards
- **Maps:** Leaflet + OpenStreetMap for live vehicle tracking
- **ML Engine:** Python for predictive maintenance & fuel anomaly detection

Vehicle Management & Predictive Insights



Vehicle & Trip Management

- 100 vehicle records (Trucks, Vans, Buses) for demonstration
- Real-time status tracking: **Active, Idle, Maintenance**
- Trip management with routes, distances, and duration
- Complete vehicle maintenance history and service records

Predictive Maintenance with ML

- Analyzes vehicle usage patterns and service history
- Identifies fuel consumption anomalies
- Predicts maintenance needs before failures occur
- Factors: Vehicle age, mileage, maintenance records, fuel patterns



Predictions are system-generated insights based on simulated project data.

Reports, Benefits & Future Scope

1

Available Reports

- Fleet Summary & Vehicle Performance
- Maintenance History
- Fuel Consumption & Cost Analysis
- Trip Analytics

2

Business Benefits

- Better operational visibility & real-time monitoring
- Reduced manual work and administrative overhead
- Improved maintenance planning and cost control
- Optimized fuel management and reduced waste
- Data-driven decision making

3

Future Enhancements

- Real GPS integration & IoT sensors
- WebSocket real-time updates
- Advanced ML models & Driver management
- Role-based access & Cloud deployment



Thank You

Fleet Management Dashboard – Transportation Domain

G. KANIMOZHI | E25CA147 | BCA Sec "A"